

Total Questions: 35

LEVEL – II

SET - B

Time: 1.5 hr.
Total Marks: 70

SECTION - 1

- This section contains **FIFTEEN (15)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 2

- This section contains **EIGHT (08)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each correct answer you will get **TWO** marks, if all the options are correct. If the correct answers chosen are less than the actual correct answers, partial marking (basis on weightage) will be given.
- There are **NO NEGATIVE** marks for wrong answers.

SECTION - 3

- This section contains **NINE (09)** questions of **PASSAGE TYPE OR COMPREHENSION TYPE**.
- Under each paragraph there will be three questions that has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 4

- This section contains **THREE (03)** questions. These questions are **MATCHING TYPE BASED QUESTIONS**.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

Section - 1

1. A farmer models the area of a rectangular field as $(x + 3)(x + 5)$, where x is length in meters. What is the expanded form representing total area?

- (A) $x^2 + 8x + 15$ (B) $x^2 + 15$
(C) $x^2 + 5x + 3$ (D) $x^2 + 8x$

2. A square park has area 2500 m^2 . A person walks along one side. How far does he walk?



- (A) 40 m (B) 45 m
(C) 50 m (D) 60 m
3. A car covers a certain distance in 8 hours at a fixed speed. If the speed is doubled, how much time will it take to cover the same distance?
(A) 2 (B) 4
(C) 8 (D) 16
4. A solid sphere of radius 7 cm is melted and recast into identical cones of radius 3.5 cm and height 14 cm. Number of cones formed is _____
(A) 2 (B) 3
(C) 4 (D) 8
5. A student saves 20% of her monthly pocket money. If her pocket money increases by 25%, how much percent does her savings increase (assuming same saving habit)?
(A) 20% (B) 25%
(C) 30% (D) 50%
6. A data server stores files in compressed form. A file of size 10^8 bytes is reduced by dividing twice by 10^2 . What is the final size?
(A) 10^4 (B) 10^6
(C) 10^8 (D) 10^{10}
7. A student observes a circular table from directly above. What shape will he see?
(A) Rectangle (B) Circle
(C) Triangle (D) Square

8. A recipe uses $\frac{2}{3}$ cup sugar, but only $\frac{3}{5}$ of that amount is used. Find actual sugar used.

(A) $\frac{2}{5}$ (B) $\frac{1}{2}$
(C) $\frac{2}{3}$ (D) $\frac{1}{5}$

9. A company paints the outer curved surface of a conical storage tank (radius = 7 m, slant height = 10 m). They need to estimate the area to be painted. find the curved surface area.



- (A) 220 m^2 (B) 440 m^2
(C) 154 m^2 (D) 308 m^2
10. A two-digit number has digits whose sum is 12. The number is divisible by 4. Which of the following could be the number?
(A) 48 (B) 83
(C) 66 (D) 57
11. The area of a square garden increases from x^2 to $(x + 2)^2$. What is the increase in area?
(A) $2x + 2$ (B) $4x + 4$
(C) $x^2 + 4$ (D) $2x + 4$
12. A shopkeeper gives a discount such that the final price of an item becomes ₹840. If the original price was increased by ₹200 and then discounted by ₹160 to reach the same final price, what is the original price?
(A) ₹800 (B) ₹960
(C) ₹1000 (D) ₹840

13. A cube-shaped box is examined. How many edges does it have?



- (A) 8 (B) 10
(C) 12 (D) 6
14. A person walks 6 km in 2 hours. How much distance will be covered in 5 hours at the same speed?
- (A) 10 (B) 12
(C) 15 (D) 18
15. A trader buys 50 items at ₹80 each. He sells 40 items at ₹100 each and the remaining at ₹60 each. Find his overall profit/loss percentage.
- (A) 15% profit (B) 5% loss
(C) 2% profit (D) 2% loss

Section - 2

16. A student claims that dividing any two rational numbers always gives a rational number. Another student gives example: $5 \div 0$ and says it is undefined. Which conclusions are correct?
- (A) Rational numbers are closed under division
(B) Rational numbers are not closed under division
(C) Division by zero is not defined
(D) Closure holds if denominator $\neq 0$

17. A right square pyramid has base side a and slant height l . Which are correct?

- (A) Area of one triangular face $= \frac{1}{2}al$
(B) Total lateral area $= 4al$
(C) Number of triangular faces $= 4$
(D) Total lateral area $= 4 \times \frac{1}{2}al$

18. A number leaves remainder 2 when divided by 5 and remainder 3 when divided by 4. Students try to find such numbers less than 50. Which are correct?

- (A) 7 (B) 27
(C) 47 (D) 31

19. A system follows: $10^5 \div 10^3 \times 10^{-2}$. Which statements are correct?

- (A) Final answer is 10^0
(B) Quotient rule is used
(C) Product rule is used
(D) Final answer is 10^4

20. A shopkeeper gives successive discounts of 10% and 20%. Which of the following statements are correct?

- (A) Equivalent discount is 30%
(B) Equivalent discount is 28%
(C) Customer pays 72% of marked price
(D) Customer pays 70% of marked price

21. Which of the following situations can be represented by the equation

$$x + 15 = 45?$$

- (A) A number increased by 15 gives 45
(B) A number decreased by 15 gives 45
(C) Sum of a number and 15 is 45
(D) A number multiplied by 15 gives 45

22. Which of the following numbers are perfect squares?

- (A) 441 (B) 450
(C) 529 (D) 540

23. A scientist uses the expression $4x^4 + 2x^2$ in modeling. Which are true?

- (A) It is bi-quadratic
- (B) Highest degree is 4
- (C) It is cubic
- (D) It has two terms

Section - 3

PASSAGE 1:

A cuboidal water tank has dimensions $10\text{ m} \times 6\text{ m} \times 5\text{ m}$. The tank is filled with water up to full capacity. The inside walls and base are coated with a waterproof layer.



24. What is the total area to be coated?

- (A) 220 m^2
- (B) 260 m^2
- (C) 280 m^2
- (D) 300 m^2

25. How much water (in m^3) can the tank hold?

- (A) 250
- (B) 300
- (C) 350
- (D) 400

26. If water is filled at $2\text{ m}^3/\text{min}$, how long will it take to fill the tank?

- (A) 120 min
- (B) 150 min
- (C) 180 min
- (D) 200 min

PASSAGE-2

A school collects money for a science exhibition. The total collection is ₹4800. The amount collected from senior students

is twice the amount collected from junior students. Additionally, ₹600 is donated by teachers.

Let the amount collected from junior students be ₹x.

27. Which equation correctly represents the situation?

- (A) $x + 2x + 600 = 4800$
- (B) $2x + 600 = 4800$
- (C) $x + 2x = 4800$
- (D) $3x - 600 = 4800$

28. The amount collected from junior students is:

- (A) ₹1400
- (B) ₹1200
- (C) ₹1000
- (D) ₹1600

29. Amount collected from senior students is:

- (A) ₹2400
- (B) ₹2800
- (C) ₹2600
- (D) ₹3000

PASSAGE-3

A city designer plans a square plaza with **2025 tiles** arranged equally in rows and columns.

He divides it into **smaller square sections** of equal tiles.

Later, he adds tiles to make the total **2500 tiles**.

He checks if both arrangements form perfect squares.

He also studies patterns using square roots and factorisation.

30. The number of tiles in each row of the initial plaza is:

- (A) 40
- (B) 45
- (C) 50
- (D) 55

31. After expansion, the number of tiles in each row becomes:

- (A) 45 (B) 48
(C) 50 (D) 55

32. How many tiles were added to expand the plaza?

- (A) 425 (B) 450
(C) 475 (D) 500

Section - 4

33. An architect designs cube-shaped rooms and storage units. Match each situation with the correct value

Column I (Situation)	Column II (Value)
(a) Volume of a cube room with side 8 m	(i) 512
(b) A number whose cube is 343 used in scaling models	(ii) 7
(c) Cube of -4 representing depth below ground level	(iii) - 64
(d) A cube box storing $3 \times 3 \times 3$ items	(iv) 27

- (A) a-i, b-ii, c-iii, d-iv
(B) a-ii, b-i, c-iv, d-iii
(C) a-i, b-ii, c-iv, d-iii
(D) a-iii, b-ii, c-i, d-iv

34.

Column I (Situations)	Column II (Equations)
(a) A number when doubled and increased by 5 equals thrice the number decreased by 7	(i) $x - 3 = 2x + 9$
(b) Renu's age is 4 times her brother; 5 years ago it was 9 times	(ii) $7x - 4 = 5x + 10$
(c) A number reduced by 3 equals twice the number increased by 9	(iii) $2x + 5 = 3x - 7$
(d) A number multiplied by 7 and reduced by 4 equals 5 times the number increased by 10	(iv) $4x - 5 = 9(x - 5)$

- (A) a-i, b-ii, c-iii, d-iv
(B) a-iii, b-iv, c-i, d-ii
(C) a-iii, b-ii, c-iv, d-i
(D) a-ii, b-iii, c-i, d-iv

35.

Column I (Situations)	Column II (Values)
(a) If 3 items cost ₹120, find the cost of 6 items	(i) 350
(b) If 8 workers make 16 objects in the same time, find how many objects 16 workers will make	(ii) 240
(c) If 10 units are produced in 2 minutes, find how many units will be produced in 4 minutes	(iii) 32
(d) If 1 item costs ₹50, find the cost of 7 items	(iv) 20

- (A) a-i, b-ii, c-iii, d-iv
 (B) a-ii, b-iii, c-iv, d-i
 (C) a-ii, b-iii, c-i, d-iv
 (D) a-iii, b-ii, c-i, d-iv